

CHAINAT AGROMET, Thailand

WMO: 484020

Lat: 15.1581N

Lon: 100.1914E

Elev: 17

StdP: 101.12

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	18.5	11.0	8.2	21.6	12.8	9.2	22.8	6.2	25.8	5.2	25.8	0.5	N/A	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.9	37.6	26.0	36.4	26.2	35.4	26.1	28.0	33.3	27.6	32.9	27.3	32.5	1.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.4	30.3	26.4	21.9	30.0	26.1	21.5	29.7	89.9	33.5	88.1	33.1	86.7	32.6	33.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.5	4.6	3.5	DB	15.1	39.0	1.9	1.3	13.7	40.0	12.6	40.7	11.5	41.5	10.1	42.5	
			WB	12.7	29.8	1.6	1.3	11.5	30.7	10.6	31.5	9.6	32.2	8.4	33.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	25.8	27.7	29.5	30.9	30.4	29.7	29.0	28.9	28.7	28.5	27.6	26.0
	DBStd	2.20	2.07	1.74	1.93	1.57	1.70	1.11	1.08	0.95	1.02	1.14	1.80	2.19
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6776	488	496	604	628	633	590	589	586	562	575	529	496
	CDD18.3	3735	231	262	347	378	375	340	331	327	312	317	279	237
	CDH23.3	42316	2027	2719	4162	5017	4799	4142	3789	3653	3424	3526	2904	2153
Wind	CDH26.7	18833	777	1197	2078	2770	2453	1872	1552	1437	1329	1377	1158	832
	WSAvg	0.7	0.5	0.6	0.9	1.0	0.9	1.0	0.9	0.8	0.5	0.4	0.6	0.7
	PrecAvg	1161	8	10	43	74	150	132	138	176	255	144	26	6
	PrecMax	1506	35	53	136	164	258	198	245	289	381	414	104	46
	PrecMin	805	0	0	0	10	33	41	58	88	155	8	0	0
	PrecStd	168	9	12	33	44	56	45	44	44	60	75	28	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	35.2	38.0	39.5	38.9	36.6	35.5	34.7	34.6	34.4	34.6	34.2
		MCWB	24.2	25.2	24.2	25.3	26.6	26.5	26.5	26.7	27.2	26.2	25.6	24.8
	2%	DB	32.7	34.1	36.2	37.9	37.2	35.2	34.2	33.7	33.6	33.4	33.6	33.1
		MCWB	23.8	25.2	25.4	25.8	26.8	26.5	26.4	26.5	26.8	26.1	25.1	24.6
	5%	DB	31.8	33.3	35.2	36.7	35.9	34.1	33.2	33.0	32.9	32.7	32.8	32.1
		MCWB	23.6	25.0	25.5	26.1	26.7	26.6	26.3	26.3	26.6	26.0	24.9	23.8
	10%	DB	30.8	32.4	34.3	35.6	34.7	33.1	32.3	32.1	32.0	32.0	31.9	31.0
		MCWB	23.1	24.7	25.5	26.1	26.7	26.5	26.2	26.2	26.4	25.8	24.6	23.1
	0.4%	WB	26.5	27.2	27.5	28.2	28.6	28.1	27.9	27.8	28.3	27.7	27.5	26.6
		MCDB	31.1	32.8	33.5	34.7	34.6	32.8	32.6	32.5	32.5	31.8	31.3	31.5
	2%	WB	25.7	26.6	27.0	27.7	27.8	27.5	27.2	27.3	27.6	27.3	26.8	25.7
		MCDB	30.3	32.1	33.2	33.9	33.9	32.5	32.2	32.0	31.8	31.3	31.2	30.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	25.0	26.2	26.6	27.2	27.4	27.1	26.8	26.9	27.1	26.9	26.3	25.1
		MCDB	29.5	31.5	32.6	33.3	33.3	32.3	31.7	31.6	31.4	30.9	30.4	29.7
	10%	WB	24.4	25.7	26.2	26.8	27.1	26.7	26.5	26.5	26.7	26.6	25.7	24.4
		MCDB	28.6	30.7	32.0	32.8	32.8	31.7	31.1	30.9	30.8	30.3	29.5	29.0
	5% DB	MDBR	10.4	10.0	10.0	9.9	8.4	7.4	7.0	7.0	7.0	6.9	8.4	10.0
		MCDBR	11.2	10.7	11.1	11.0	10.0	8.7	8.3	8.2	8.0	7.9	9.2	10.5
Mean Daily Temperature Range	5% WB	MCWBR	4.0	3.5	2.9	2.7	2.6	2.8	2.8	2.8	2.8	2.6	2.7	3.4
		MCDBR	9.3	9.7	10.1	10.0	9.1	7.8	7.5	7.6	7.6	7.2	8.2	9.4
	5% WB	MCWBR	3.9	3.5	3.2	3.1	3.0	2.8	2.9	3.0	2.9	2.8	3.2	3.5
Clear-Sky Solar Irradiance	taub		0.520	0.605	0.669	0.597	0.499	0.447	0.438	0.441	0.477	0.565	0.505	0.505
	taud		1.982	1.806	1.693	1.896	2.221	2.370	2.394	2.372	2.259	1.967	2.108	2.040
	Ebn at Noon		772	725	693	745	809	844	853	859	832	747	781	772
	Edh at Noon		175	217	249	203	145	123	121	125	139	182	153	161
All-Sky Solar Radiation	RadAvg		4.80	5.26	5.50	5.99	5.63	5.28	4.97	4.88	4.81	4.79	4.84	4.70
	RadStd		0.29	0.21	0.36	0.29	0.37	0.32	0.28	0.31	0.26	0.28	0.27	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	+0.19	+0.24	N/A	N/A	+93	+91

Nomenclature: See separate page